ANNISA SELVIRA RA'INA GUSTIYANINGSIH

Electrical Engineer | Entrepreneur | Operations & Control Systems Specialist
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PROFESSIONAL SUMMARY

Award-winning Electrical Engineer (Universitas Telkom, GPA 3.53/4.00) led a 6-person team to 1st Place at KRI 2023 Regional 1 and authored a peer-reviewed publication on hybrid power generation for remote island electrification. Founder and active Operations Manager of Jasa Transport Selwi (Mar 2023 – Present), a freight logistics business with 7 vehicles, 7 drivers, and 1,000+ clients across Central Kalimantan demonstrating direct P&L ownership, resource allocation, SOP development, and team leadership. Specialized in control systems automation (PLC Omron, MATLAB/Simulink), embedded systems (Arduino, STM32, PCB Design), and renewable energy simulation (HOMER, DIgSILENT, QBlade), with advanced Python for data analysis and machine learning. Committed to driving operational excellence and measurable technical impact in manufacturing, energy, or industrial technology environments.

TECHNICAL SKILLS

Control & Automation: PLC Omron (Advanced), MATLAB/Simulink (Advanced), DIgSILENT PowerFactory, HOMER, SCADA Systems, Proteus/PSPICE

Embedded Systems & Hardware: Arduino Uno/Mega (Advanced), STM32, NodeMCU ESP8266, PCB Design SMD/Non-SMD — Autodesk Eagle (Advanced), Sensor & Actuator Integration

Programming: C/C++ (Advanced), Python (Advanced — Pandas, NumPy, Scikit-learn, Matplotlib, subprocess, os), MySQL (Intermediate), Java, C#

Data Science & AI: Scikit-learn, EDA, Supervised ML — Classification & Regression, Confusion Matrix Analysis, Feature Engineering

Renewable Energy: QBlade Wind Turbine Blade Design, PVsyst Solar Simulation, HOMER Hybrid System Optimization, On-Grid & Off-Grid Solar PV Design

Design & Modeling: SolidWorks, Autodesk Inventor, Autodesk Eagle, Blender 3D

Industry Tools: Microsoft Office — Excel, PowerPoint, Word (High Proficiency), Figma UI/UX, Git Version Control

Operations & Leadership: Business Operations Management, P&L Oversight, Fleet & Resource Allocation, KPI Monitoring, SOP Development & Optimization, Team Leadership, Curriculum Development, Stakeholder & Client Communication, Partnership Development, Risk Analysis, Financial Modeling

Professional Skills: High Initiative, Critical Thinking, Independent Decision Making, Problem-Solving, Adaptability, Cross-functional Collaboration, Technical Mentoring, Public Speaking

Languages: Bahasa Indonesia (Native), English (Professional Working Proficiency)

ENTREPRENEURSHIP

Founder & Operations Manager Mar 2023 – Present

Jasa Transport Selwi | Kalimantan Tengah

Founded and actively managing a B2B freight and logistics transport business serving inter-city routes across Central Kalimantan, with full P&L ownership and end-to-end operational accountability.

- Built and scaled a freight logistics operation from zero to a fleet of 7 vehicles (5 pick-up trucks + 2 trucks) and a team of 7 drivers within 3 years, achieving a stable client base of 1,000+ across multiple inter-city routes in Central Kalimantan.
- Manage full daily operations including route scheduling, driver deployment, fleet maintenance planning, and shipment monitoring sustaining consistent on-time delivery performance and zero critical service failures across active client accounts.
- Developed and maintained a 1,000+ client base through direct relationship management, service quality consistency, and proactive issue resolution sustaining stable monthly revenue growth with minimal client churn year-over-year.
- Oversee end-to-end business financials including operational cost control, driver payroll management, vehicle maintenance budgeting, and monthly P&L monitoring to ensure sustained business profitability and cash flow stability.
- Designed and implemented operational SOPs for driver coordination, cargo handling, and client communication — improving service consistency and enabling smooth business continuity during disruptions such as vehicle breakdowns or route changes.

- Proactively identified and resolved operational disruptions (breakdowns, route delays, driver performance issues) independently without third-party escalation, protecting client trust and maintaining service reliability across all active routes.

PROFESSIONAL EXPERIENCE

Branch Coordinator & Teacher

May 2024 – Mar 2026

Coding Bee Academy | Kab. Bandung Barat, Jawa Barat

Led full branch operations and technology education delivery for a growing STEM academy, applying project management, KPI monitoring, SOP development, and team leadership skills directly transferable to engineering and operations management roles.

Branch Coordinator

- Managed end-to-end branch operations overseeing 5+ educators and 50+ active students across concurrent programs sustaining consistent quality delivery and measurable academic outcomes aligned with organizational KPIs and management reporting requirements.
- Secured partnerships with 3+ local schools and community organizations through structured outreach campaigns and formal institutional presentations, contributing directly to a 20% increase in student enrollment within 12 months.
- Elevated educator performance through structured monthly coaching and professional development sessions, improving curriculum delivery consistency and reducing staff turnover across all teaching cohorts.
- Implemented structured enrollment tracking and parent communication systems, sustaining student retention above 85% and maintaining high program satisfaction scores across all cohorts throughout the operational period.
- Contributed to branch strategic expansion by identifying market opportunities, developing formal partnership proposals, and providing structured operational performance reports directly to senior management.

Teacher

- Designed and delivered project-based technology curricula across 6 subject areas including Robotic Engineering (microcontrollers, sensors, automation) and Back-End Web Development (REST APIs, databases) — increasing student project completion rates by approximately 25% through active hands-on learning methodologies.
- Organised and facilitated 5+ workshops, hackathons, and technology competitions annually, measurably boosting student engagement scores and real-world problem-solving capability beyond standard coursework.

Research Laboratory Assistant

Oct 2021 – Oct 2023

Electronics & Intelligence Robotics Research Group (EIRRG) | Universitas Telkom, Bandung

Supported applied robotics R&D within a university research group focused on autonomous systems, embedded hardware, and sensor integration.

- Executed PCB schematic capture and SMD layout using Autodesk Eagle across 5+ robotics prototypes, achieving zero critical layout errors and contributing to hardware validation milestones supporting publication-ready autonomous systems research.
- Streamlined component procurement tracking for 10+ component categories via a structured Excel-based monitoring system, reducing procurement lead times by an estimated 30% and improving R&D budget transparency.
- Assisted in firmware debugging and sensor integration testing for microcontroller-based prototypes, supporting structured data collection and technical documentation for applied robotics research publications.

Engineering Intern

Jul 2022 – Sep 2022

PT Telkom Indonesia (Persero) Tbk | Kotawaringin Timur, Kalimantan Tengah

Completed a 3-month internship at a regional telecom infrastructure facility, gaining direct exposure to industrial power systems in a live, mission-critical production environment.

- Analysed 10+ ATS (Automatic Transfer Switch) panel diagrams, single-line diagrams, and live voltage flow data identifying key power continuity dependencies across PLN and backup generator systems, with findings directly informing team maintenance strategy and contingency planning.
- Documented relay protection configurations and device settings for 5+ panel systems, producing structured technical reports that supported team knowledge transfer, system maintenance planning, and operational continuity assurance.

KEY PROJECTS

Multi-Sensor Industrial Fire Safety System Dec 2025 – Feb 2026

Personal Project

- Engineered a standalone Arduino-based industrial fire safety system integrating dual flame sensors and an analog temperature sensor with dual-stage risk logic (Warning: >40°C/single flame; Danger: >60°C/dual flame/emergency trigger), a safety-latching alarm mechanism (1000 Hz buzzer + LED indicator), and a secure multi-condition reset interlock fully designed, programmed, and validated on physical hardware prototype.

Hybrid Power Generation Feasibility Study — Karimunjawa Island Aug 2022 – Aug 2023

Capstone Research | Universitas Telkom

- Led techno-economic analysis of 4 hybrid power generation configurations; identified PLTD+Wind as optimal, reducing Cost of Generation from Rp1,326/kWh to Rp1,231/kWh over a 20-year projection (2023–2043). Designed 100 kW wind turbine blade (QBlade, NACA 2414 airfoil) and validated system stability under short-circuit and intermittency conditions in DIgSILENT PowerFactory resulting in a peer-reviewed published research paper on remote island electrification.

Machine Learning Handwritten Letter Recognition (EMNIST) Sep 2022 – Feb 2023

ML & Applications Course | Universitas Telkom

- Built an end-to-end OCR-style multi-class classification pipeline using Scikit-learn, Pandas, and NumPy on the 145,600-sample EMNIST Letters dataset covering preprocessing, feature engineering, EDA, supervised classification, and confusion matrix evaluation, achieving reliable recognition across all 26 alphabet classes.

Solar PV On-Grid System Design & Economic Analysis Sep 2022 – Feb 2023

Renewable Energy Economics Course | Universitas Telkom

- Designed a residential on-grid PLTS system for Jakarta using PVsyst, achieving 6.40 MWh/year estimated output at 79.42% performance ratio. Conducted full techno-economic evaluation (LCC, LCOE, NPV, Profitability Index, Payback Period) confirming investment feasibility and quantifying CO2 reduction impact for sustainable energy implementation.

EDUCATION

Bachelor of Engineering — Electrical Engineering Aug 2019 – Aug 2023

Universitas Telkom | Bandung, Jawa Barat

- GPA: 3.53 / 4.00 | Specialisation: Control Systems, Power Systems, Automation & Smart Energy Technologies.
- Capstone Project: Hybrid renewable energy feasibility study for remote island electrification resulted in a peer-reviewed published research paper on wind turbine blade design and power system cost optimization.
- Relevant Coursework: Digital Control Systems, Power Electronics, Embedded Systems, Renewable Energy Economics, Machine Learning & AI Applications, IoT Engineering, Robotics Engineering, PCB Design & Fabrication.
- Academic Achievement: Selected to represent Universitas Telkom at Indonesian Robot Contest (KRI 2023) awarded 1st Place, Regional 1, KRSRI Division (Kontes Robot Seni Budaya Indonesia).

ACHIEVEMENTS & PUBLICATIONS

- 1st Place Regional 1, Indonesian Robot Contest (KRI 2023), KRSRI Division (Kontes Robot Seni Budaya Indonesia). Led 4-person interdisciplinary team designing and building a fully autonomous rescue robot capable of navigating earthquake and landslide simulation environments and detecting victims competing against university teams across Regional 1.
- Peer-Reviewed Publication: "Wind Turbine Design for a Power Generation System with Competitive Generation Costs on Karimunjawa Island" 100 kW aerodynamic blade design (NACA 2414 airfoil) using QBlade, validated via DIgSILENT PowerFactory and HOMER, contributing to renewable electrification research for remote Indonesian islands.

CERTIFICATIONS & TRAINING

- Fundamental AI & Coding Training (3-month program, Completed: May 2025) — Kementerian Pendidikan Dasar dan Menengah. Covered AI fundamentals, coding implementation, and technology integration in education.

- Python for Data Professional: Beginner → Visualization → EDA (COVID-19 Dataset Analysis) — DQLab Bootcamp (Completed: Aug 2021)
- Exploratory Data Analysis with Python & Pandas | Data Visualization with Matplotlib | Fundamental SQL Using SELECT Statement | R Fundamental for Data Science — DQLab Bootcamp (Completed: Aug 2021)